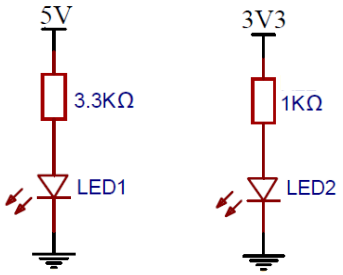
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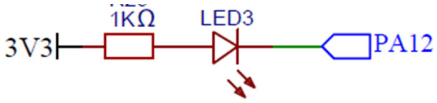
SC2104/CE3002 Sensors & Interfacing
Tutorial 1 – Interfacing Circuits

1. The following schematic diagrams are of two circuits used on the STM32F04 embedded board.



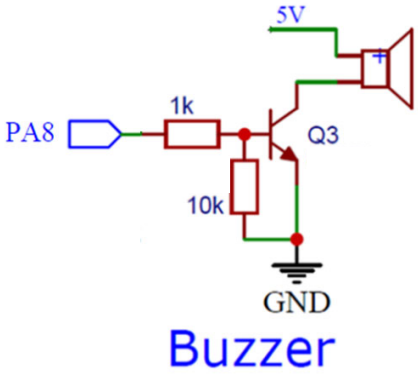
With reference to information given in lecture note Topic 1, estimate the current that is flowing through the each of the two LEDs when they are turned on.

2. The following circuit is from the same embedded board where LED3 is driven through pin 12 of port A (PA12)



Explain whether PA12 should be configured as an input or output pin, and its appropriate setting (e.g. push-pull, open-collector) in order to operate the LED3.

3. The following schematic diagram shows the buzzer circuit used on the embedded board.

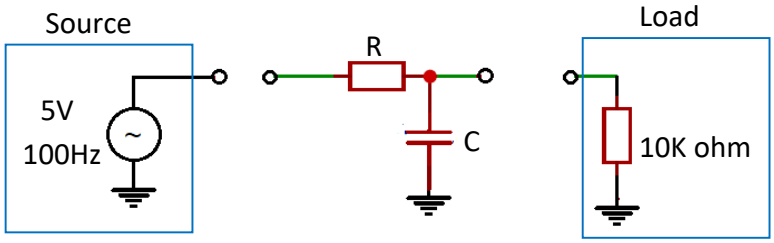


Given that

- the output high voltage level of PA8 from the microcontroller is 3.3V
- the current gain of the transistor Q3 is 10
- voltage drop across Base-Emitter terminal of Q3 is 0.7V when conducting

Find the current flowing through the buzzer and check that it is operating below its maximum rated current of 50mA.

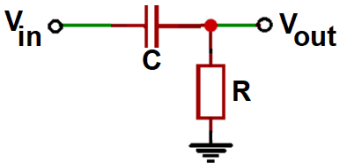
4. A Low Pass RC filter is to be designed as shown in the following diagram such that it has a corner frequency of 1KHz, to be used with a signal source that contains 5V 100 Hz signal (which is corrupted with a 15KHz noise).



Furthermore, output of the RC Lowpass filter is to be connected to a circuit (known as the 'load') that has an equivalent input impedance of 10K ohm as shown in the diagram.

Calculate the R and C values that you will use for this circuit.

5. By swapping the position of the two components of a low pass filter as shown below, the circuit becomes a high pass filter.



a. Explain, in an intuitive way, how this circuit will exhibit the characteristics of a high pass filter.

b. Derive the expression of the -3dB corner frequency of this high pass filter.